

Software Project Management assignment

Participants:

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Supply Gate

Supply Gate is a trusted online marketplace designed to connect businesses with verified manufacturers and suppliers. It provides a secure platform where companies can easily search, compare, and engage with reliable partners. Through supplier verification, businesses gain confidence in the authenticity and quality of products and services. The platform reduces the time and cost involved in sourcing while promoting transparency in trade. By bridging the gap between buyers and suppliers, Supply Gate enhances business growth and fosters long-term partnerships.

Business Objectives

- Establish a secure and reliable online marketplace that connects businesses with verified manufacturers and suppliers.
- Successfully onboard at least 100 verified suppliers within the first six months of launch.
- Achieve and maintain a minimum 90% satisfaction rating from business users within the first year.
- Generate significant transaction volume (target: \$15,000) during the first year of operation.
- Streamline the supplier search process by reducing the time businesses spend finding suitable partners by at least 60%.

Current Situation

- Businesses often face challenges in identifying reliable suppliers due to a lack of verification, limited transparency, and time-consuming search processes. Existing platforms may not provide sufficient trust mechanisms, as supplier authenticity is not always guaranteed. Moreover, small and medium enterprises struggle to

access affordable, streamlined systems that connect them with qualified manufacturers.

Problem/Opportunity Statement

- **Problem statement** is the absence of a trusted, verified supplier marketplace leads to inefficiencies, delays, and potential risks in procurement. Businesses waste valuable time evaluating suppliers manually and risk engaging with unverified or fraudulent vendors.
- **The opportunity** lies in developing **Supply Gate** — a centralized, trusted online marketplace that ensures supplier verification, simplifies supplier discovery, and reduces procurement time. This platform not only enhances trust between businesses and suppliers but also creates a scalable ecosystem that supports long-term partnerships and business growth.

Critical Assumptions

These are things we have that are critical so project will work:

Critical Conditions

1. **Supplier Verification**
 - Enough manufacturers and suppliers provide verifiable information for onboarding.
2. **Business Adoption**
 - Businesses are willing to use a new platform for sourcing suppliers.
3. **Platform Usability**
 - Users (both businesses and suppliers) can navigate the platform without extensive training.
4. **Internet & Tech Accessibility**
 - Target users have access to reliable internet and compatible devices.
5. **Regulatory Compliance**
 - No legal restrictions prevent collecting supplier data or facilitating B2B transactions in the target regions.

Constraints

1. **Budget**
 - Limited funding affects marketing, tech infrastructure, or the speed of onboarding suppliers.
2. **Time**
 - Onboarding 100 verified suppliers within 6 months and achieving target satisfaction ratings imposes a strict timeline.
3. **Technology Stack**
 - The platform is limited to certain programming languages, frameworks, or third-party integrations.

4. **Data Security & Privacy**
 - Must comply with data protection regulations (e.g., GDPR, local laws), limiting data storage or handling options.
5. **Market Limitations**
 - Some regions have fewer suppliers or businesses ready to adopt online marketplaces.

Analysis of Options & Recommendation

- **Option 1: Outsource Development** – Fast delivery, higher cost, limited control.
 - **Option 2: Develop In-House** – Cost-effective, full control, easier to adapt platform features.
 - **Recommendation:** Build in-house using open-source tools and team expertise to maintain control, minimize costs, and allow rapid feature updates.
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Needs Assessment & Business Case Details

Core Features of South Gate Platform

- Verified supplier onboarding and business account management
- Searchable supplier directory with filters and ratings
- Secure messaging and transaction tracking
- Admin dashboard for usage analytics
- Data encryption and secure login

Budget Estimate and Financial Analysis

Budget	Estimate
Development & Hosting	\$720
Design & UX	\$480
Testing	\$240
Contingency	\$360
Total	\$1800

Schedule	Estimate
Planning & Requirements	2 weeks
Design	2 weeks
Development	1 Month
Testing & Review	1 Month
Total	3 Months

Potential Risks

Risk	Likelihood	Impact	Mitigation
Delay in Development	Medium	High	Weekly stand-ups, tracking
Data security or privacy	Low	High	Strong encryption, Secure login
Technical issue / downtime	Medium	Medium	Regular testing ,backup system
Budget overrun	Low	High	Monitor expense

Key Exhibits / Deliverables – South Gate

1. **Verified Supplier Database** – Demo data and sample supplier profiles
2. **Business Account Portal** – **Wireframes or mockups** of main interface (Figma /Adobe XD)
3. **Transaction Management System** – Sample transaction workflow or test data
4. **Admin Dashboard** – **Gantt chart** for project schedule and reporting mockups
5. Secure Login & Data Privacy Features .

2. For Identifying and Engaging Stakeholders

Stakeholder Name	Contact Information	Project Role	Project Requirement	Project Concerns	Imp influ
RUTAGANDA J Valente	+2507999999023 Valante@gmail.com	Product Manager	Fully functioning app with all high priority	Missed deadlines, feature incompleteness	High High
NDIKUMANA Norbert	+25079909090 norbert@gmail.com	Project Manager	Completion to scope,schedule and budget	Scope creep, budget overruns	High High
KAYIRANGA Cedrick	+250790345212 Cedro@gmail.com	PMO Representative	Alignment with Strategy and Integration	Strategic misalignment, lack of coordination	Med High
RUSIBANA PAUL	+250785462585 paul@gmail.com	Development Team	Delight the users, gain team lead experience	Poor UI/UX, limited innovation, lack of mentoring support	High Med
MANZI Ezra	+250794509213 ezra@gmail.com	Sponsor	Compelling UI,fast and responsive	Poor app performance unattractive UI	High High

Stakeholder Mapping (Power vs. Interest Grid)



SOFTWARE PROJECT MANAGEMENT ASSIGNMENT

Project: Supply Gate – B2B Marketplace Platform

3. RACI Chart

Task / Deliverable	Project Manager (Norbert)	Product Manager (Valente)	PMO Rep (Cedrick)	Developer (Paul)	Sponsor (Ezra)
Project Planning & Initiation	A	R	C	I	I
Requirements Gathering	R	A	C	C	I
UI/UX Design Approval	C	A	I	R	C
Platform Development	I	C	I	A/R	I
Testing & Quality Assurance	C	C	I	A/R	I
Supplier Onboarding Strategy	C	A	C	I	R
Risk Management	A	C	R	I	I
Budget & Financial Control	A	C	R	I	C
Stakeholder Communication	A	C	R	I	C
Final Review & Launch	A	R	C	R	A

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

4. Team Charter

Shared Values

Integrity: Operate transparently and uphold honesty in all project dealings.

Accountability: Every team member owns their responsibilities and meets commitments.

Collaboration: Work collectively toward shared goals with mutual respect.

Innovation: Encourage creativity and continuous improvement throughout development.

Behavior Guidelines

Respect deadlines and meeting times.

Communicate clearly and professionally.

Support and mentor one another to ensure overall success.

Maintain a positive and solution-oriented attitude.

Communication and Tool Usage

Primary Communication: WhatsApp and Google Meet.

Document Sharing: Google Drive.

Formal Communication: Email for official updates.

Response Time: Within 24 hours on working days.

Decision-Making Guidelines

Decisions are made by consensus when possible.

If conflict arises, the Project Manager decides after consulting stakeholders.

All key decisions must be documented and communicated.

Performance Expectations

Deliver high-quality outputs within agreed timelines.

Actively participate in meetings and provide updates.

Seek feedback proactively and continuously improve.

Conflict Resolution Measures

Address issues respectfully within the team.

Escalate unresolved matters to the Project Manager.

Engage Sponsor mediation when internal solutions fail.

Meeting Frequency and Channel

Weekly Team Meeting: Monday 10:00 AM via Google Meet.

Progress Reviews: Bi-weekly on Fridays.

Ad hoc Meetings: As needed with 24-hour notice.

Other Team Agreements

Shared working hours: 9:00 AM – 5:00 PM.

Each member maintains documentation in shared folders.

Any planned absence reported 2 days in advance.

5. Project Charter: Supply Gate – B2B Marketplace Platform

Project Manager: NDIKUMANA Norbert

Sponsor: MANZI Ezra

Date: 5th October 2025

Project Vision

To create a secure, reliable, and scalable B2B online marketplace that connects verified suppliers with businesses seeking trusted sourcing partners.

Project Goals

Launch a verified supplier directory with secure onboarding.

Enable businesses to source products efficiently and safely.

Build user trust through data verification and secure transactions.

Achieve \$15,000 in transactions within the first year.

Known Challenges

Challenge	Mitigation Strategy
Limited funding for marketing	Begin with low-cost digital campaigns.
Supplier verification delays	Automate verification and maintain clear standards.
Platform security concerns	Use SSL and regular security audits.
User adoption	Offer tutorials and responsive support.

Budget / Financing

Category	Estimated Cost (USD)	Notes
Development & Hosting	\$720	Backend, APIs, and servers
Design & UX	\$480	User interface and prototype design
Testing & QA	\$240	Manual and automated tests
Contingency	\$360	Unforeseen issues
Total	\$1,800	

6. Product Scope

Purpose: Facilitate transparent and verified trade between businesses and suppliers.

Core Features

- Supplier Verification & Business Registration
- Searchable Supplier Directory with Filters
- Secure Messaging & Transaction Tracking
- Admin Dashboard for Analytics
- Secure Login & Data Encryption
- Product Listings & Catalog Management
- Review and Rating System
- Quotation Request Feature (RFQ)
- Order Management & History
- Notifications and Alerts System

2 Requirement Traceability Matrics

ID	REQUIREMENT DESCRIPTION	TYPE	SOURCE	PRI ORI TY (Mo scW)	STATUS
RQ-01	Users shall be able to register as either a "Buyer" or "Supplier"	Functional	Busines need	Mus t hav e	Approv ed
RQ-02	The system shall allow suppliers to	Functiona	Busines need	Mus t	Approv ed

	upload business licenses and certifications for verification.			have	
RQ-03	Administrators shall be able to approve or reject supplier verification requests.	Functional	Business need	Must have	Approved
RQ-04	Buyers shall be able to filter search results by supplier verification status.	Functional	User story	Should have	Approved
RQ-05	The system shall display a "Verified" badge on the profiles and listings of approved suppliers.	Functional	User story	Must have	Approved
RQ-06	The platform shall have a search function with filters for category, location, and keywords.	Functional	User story	Must have	Approved
RQ-07	Suppliers shall be able to create, edit, and deactivate product listings.	Functional	User story	Must have	Approved
RQ-08	The system shall encrypt all user passwords in the database.	Non-functional(scurity)	Business Need	Must have	Approved
RQ-09	All pages shall load in under 3 seconds	Non-Functional (performance)	Business Need	Should have	Approved
RQ-10	The platform shall be accessible via web browsers on desktop and mobile devices.	Non-Functional(usability)	Business Need	Must have	Approved
RQ-11	Buyers and suppliers shall be able to send and receive messages within the platform	Functional	User story	Must have	Approved
RQ-12	Buyers shall be able to create and send a Request for	Functional	User story	Could have	Approved

	Quotation (RFQ) to multiple suppliers.				
RQ-13	Suppliers shall receive a notification when a new RFQ is available.	Functional	User story	Should have	Approved
RQ-14	After an order is fulfilled, buyers shall be able to rate and review the supplier on a 1-5 star scale.	Functional	User story	Should have	Approved
RQ-15	The system shall generate a unique, public profile page for each supplier.	Functional	User story	Must have	Approved
RQ-16	The admin console shall display metrics for total users, verified suppliers, and active listings.	Functional	Stakeholder	Could have	Approved
RQ-17	The platform shall maintain 99.9% uptime excluding scheduled maintenance.	Non-functional(reliability)	Business need	Must have	Approved
RQ-18	User data shall not be shared with third parties without explicit consent.	Non-functional(usage)	Legal	Must have	Approved
RQ-19	User data shall not be shared with third parties without explicit consent.	Non-Functional(usage)	Stakeholder	Won't have	Approved

Acceptance Criteria

- A user can successfully complete registration and log in.
- An administrator can view a supplier's documents and change their status to "Verified."
- A buyer's search for "electronic components" with a "Verified Only" filter returns only verified suppliers in that category.

- A message sent from a buyer's account appears instantly in the supplier's message inbox.
- The platform homepage loads completely in less than 3 seconds on a standard 4G connection.

Key Deliverables

- Fully functional and tested Supply Gate web application.
- User Documentation (Help Center articles for buyers and suppliers).
- Administrator Training Manual and Session.
- Deployed Application on a live server (Hosting Environment).
- Source Code and Technical Design Documentation.

Out-of-Scope Items (Project Boundaries)

Payment Processing: The platform will facilitate connections and quotes but will not process financial transactions or handle escrow services.

- Order Fulfillment & Logistics: The platform is not responsible for shipping, delivery, or customs clearance of goods.
- Contract Law & Legal Enforcement: While the platform may provide template agreements, it does not act as a legal arbiter in disputes.
- Multi-language Support: The initial release will be in English only.
- Mobile Application: The initial release is a web-responsive site, not a native iOS or Android a

Constraints and Assumptions

Constraints:

- Budget: Not to exceed \$150,000.
- Time: Must be launched within 6 months.
- Technology: Must use cloud hosting (AWS/Azure) for scalability.

Assumptions:

- Key team members will be available for the project's duration.
- A clear definition of "verified supplier" will be provided by legal and business stakeholders.
- Users will have a stable internet connection to access the platform

Work Breakdown Structure (WBS)

1.0 Supply Gate Project

1.1 Project Management

- 1.1.1 Project Planning
- 1.1.2 Risk Management
- 1.1.3 Status Reporting

1.2 Analysis & Design

- 1.2.1 Requirements Gathering
- 1.2.2 UI/UX Design & Prototyping
- 1.2.3 System Architecture Design

1.3 Platform Development

1.3.1 User Management Module

- 1.3.1.1 Registration/Login
- 1.3.1.2 Profile Management

1.3.2 Verification Module

- 1.3.2.1 Document Upload
- 1.3.2.2 Admin Review Interface

1.3.3 Search & Listing Module

- 1.3.3.1 Search Engine
- 1.3.3.2 Supplier/Product Listing CRUD*

1.3.4 Communication Module

- 1.3.4.1 Messaging System
- 1.3.4.2 Notification Engine

1.3.5 Quotation Module (RFQ)

1.3.6 Rating & Review Module

1.3.7 Admin Console Development

1.4 Testing

- 1.4.1 Unit Testing
- 1.4.2 Integration Testing
- 1.4.3 User Acceptance Testing (UAT)

1.5 Deployment

- 1.5.1 Server Setup & Configuration
- 1.5.2 Data Migration (if any)
- 1.5.3 Go-Live Launch

1.6 Training & Documentation

- 1.6.1 User Guide Creation
- 1.6.2 Admin Training

*CRUD: Create, Read, Update, Delete

WBS ID & Name	1.3.11 Registration/Login
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description	Develop the backend and frontend components that allow a new user to create an account and an existing user to log in securely.
Responsible	Development Team
Deliverables	• Functional sign-up and login pages. • Secure password hashing mechanism. • Database schema for user credentials.
Acceptance criteria	• A new user can input details, select a role (Buyer/Supplier), and receive a confirmation email. • A registered user can log in with their email and password. • Incorrect login attempts are handled gracefully. • Passwords are stored encrypted in the database.
Assumptions & constraints	Assumption: Users have a valid email address. Constraint: Must comply with data protection regulations (e.g., GDPR).
Estimated Efforts	25 person-hours
Due date	25/september